Hypothesis Testing with E-Values

Chapter 15: Improved E-Value Thresholds under Additional Conditions

Based on Ramdas & Wang

Where we are

- **Chapters 2, 8, 11** gave us the universal safety net: for *any* e-variable E (valid under a null $\mathbb{P}$), Markov's inequality gives $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$. Rejecting when $E \geq 1/\alpha$ is always a valid level-α test, no matter how weird E's distribution is.
- That worst-case guarantee is bought at a price: if we *happen to know something extra* about the shape of E's distribution, the threshold $1/\alpha$ is needlessly conservative.
- **Chapter 15** asks: what extra, checkable, distribution-shape assumptions let us reject at a *smaller* threshold than $1/\alpha$ – giving a strictly more powerful test while remaining exactly as valid?
- Roadmap:
  - 15.1: the general machinery – the quantity $R_\gamma(\mathcal{E})$ and the resulting threshold $T_\alpha(\mathcal{E})$, for any restricted class $\mathcal{E}$ of e-variables;
  - 15.2: **comonotonic** e-variables – families that all move together;
  - 15.3: **density conditions** – decreasing or unimodal densities;
  - 15.4: conditions on the **log** of the e-variable, motivated by products of e-values (e-processes) and the CLT.
- Throughout, we fix a single (atomless) null $\mathbb{P}$; composite nulls just require the shape assumption to hold under every null distribution.

Notation you'll need I

Recap: the Markov/Ville threshold (Chapter 2, Proposition 2.1)

For *any* e-variable E (i.e. $E \geq 0$, $\mathbb{E}^{\mathbb{P}}[E] \leq 1$) and any $\alpha \in (0, 1)$,

$$\mathbb{P}(E \geq 1/\alpha) \ \leq \ \alpha.$$

So "reject H_0 iff $E \geq 1/\alpha$" is always a valid level-α test. This bound is *tight over the whole set $\mathfrak{E}$ of all e-variables* – it cannot be improved without extra information. Chapter 15's entire agenda is: restrict attention to a smaller set $\mathcal{E} \subseteq \mathfrak{E}$ satisfying some shape condition, and find the best (smallest) threshold for that smaller set.

Notation you'll need II

The two central quantities of this chapter

For $\gamma \in (0,1]$ and a set $\mathcal{E} \subseteq \mathfrak{E}$,

$$R_\gamma(\mathcal{E}) := \sup_{E \in \mathcal{E}} \mathbb{P}(E \geq 1/\gamma), \qquad T_\alpha(\mathcal{E}) := \inf\{t \geq 1 : R_{1/t}(\mathcal{E}) \leq \alpha\}.$$

Plain English: $R_\gamma(\mathcal{E})$ is the worst-case type-I error you'd get if you used the naive threshold $1/\gamma$ but restricted to variables in $\mathcal{E}$; $T_\alpha(\mathcal{E})$ is the smallest threshold that still controls type-I error at level α for every $E \in \mathcal{E}$. Always $R_\gamma(\mathcal{E}) \leq R_\gamma(\mathfrak{E}) = \gamma$, so $T_\alpha(\mathcal{E}) \leq 1/\alpha$: the restricted threshold is never worse than the naive one, and often strictly better.

Notation you'll need III

Comonotonicity, plainly (Definition 15.3)

A collection of random variables is *comonotonic* if they are all increasing functions of one common underlying random variable Z. *Analogy:* two thermometers reading the *same weather*, one in Celsius and one in Fahrenheit – different scales, but they always go up and down *together*, because both are increasing functions of the actual temperature Z. Formally: $\mathcal{X}_C$ is comonotonic if there is a common Z with every $X \in \mathcal{X}_C$ equal to $h_X(Z)$ for some increasing h_X.

Density conditions, plainly (Section 15.3)

An assumption that the e-value's distribution has a well-behaved density – ruling out weird spiky, jagged, or atomic (point-mass-heavy) distributions concentrated in unhelpful places. Two flavors we'll use: a *decreasing* density (most mass near the low end, tapering off, like Exponential or Pareto), and a *unimodal* density (rises then falls, with a single peak, like Log-normal or Gamma).

Notation you'll need IV

The log-transform idea, plainly (Section 15.4)

Many e-values in practice are *products* of many small pieces (Chapter 7's e-processes). By a Central-Limit-Theorem heuristic, $\log E$ (a sum of many small terms) should look approximately Gaussian-like: symmetric, unimodal, or at least with a well-behaved (decreasing/unimodal) density of its own. Section 15.4 asks: if we only assume something about the shape of $\log E$ (not E itself), how much can we improve the threshold?

Quantile notation (used in Lemma 15.1)

For $\beta \in (0,1)$, the left $(1-\beta)$-quantile of a random variable X is

$$q_\beta(X) := \inf\{x \in \mathbb{R} : \mathbb{P}(X \leq x) \geq 1-\beta\}.$$

Plain English: the smallest value x such that X is below x with probability at least $1-\beta$ – i.e., X exceeds x with probability at most β.

Section 15.1 – Intuition: why does restricting $\mathcal{E}$ help at all?

- Markov's inequality $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$ is proved using *only* $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ and $E \geq 0$ – it says nothing about *where* the probability mass of E sits, other than its mean.
- The worst possible e-variable (the one that actually attains $\mathbb{P}(E \geq 1/\alpha) = \alpha$) is a very particular, spiky, two-point distribution: all its mass at 0 except a point mass of size exactly α at $1/\alpha$.
- If we know E *cannot* look like that – e.g., it has a smooth, decreasing density, or is one of a comonotonic family – then this worst case is unreachable, and the true worst-case probability of exceeding a threshold is strictly smaller than the naive Markov bound suggests.
- Section 15.1 makes this precise for an *arbitrary* restricted class $\mathcal{E}$, via the quantities $R_\gamma(\mathcal{E})$ and $T_\alpha(\mathcal{E})$ introduced above; Sections 15.2–15.4 then instantiate $\mathcal{E}$ with concrete, checkable shape conditions.

15.1 Conditional e-to-p calibrators on a subset of e-values I

Definition: $R_\gamma(\mathcal{E})$

For $\gamma \in (0,1]$ and $\mathcal{E} \subseteq \mathfrak{E}$,

$$R_\gamma(\mathcal{E}) = \sup_{E \in \mathcal{E}} \mathbb{P}(E \geq 1/\gamma).$$

Since $\mathcal{E} \subseteq \mathfrak{E}$, always $R_\gamma(\mathcal{E}) \leq R_\gamma(\mathfrak{E}) = \gamma$ (Markov). We only care about $\gamma \in (0,1]$: for $\gamma > 1$ typically $R_\gamma(\mathcal{E}) = 1$ (nothing to be gained by asking about a threshold below 1, since $E \leq 1$ carries no evidence against H_0).

15.1 Conditional e-to-p calibrators on a subset of e-values II

Lemma 15.1: the optimal threshold

For $\alpha \in (0,1)$, define $T_\alpha(\mathcal{E}) := \inf\{t \geq 1 : R_{1/t}(\mathcal{E}) \leq \alpha\}$. Then

$$T_\alpha(\mathcal{E}) = \Big(\sup_{E \in \mathcal{E}} q_\alpha(E)\Big) \vee 1,$$

where $q_\alpha(E)$ is the left $(1-\alpha)$-quantile of E (previous slide) and $\vee$ denotes maximum. If $\gamma \mapsto R_\gamma(\mathcal{E})$ is continuous, $T_\alpha(\mathcal{E})$ is literally the *smallest* real number $t \geq 1$ with $\mathbb{P}(E \geq t) \leq \alpha$ for every $E \in \mathcal{E}$.

Why this is the right definition: it says "use the largest quantile that any member of $\mathcal{E}$ could produce, but never go below 1 (an e-value ≤ 1 is uninformative, so it never makes sense to reject there)."

15.1 Conditional e-to-p calibrators on a subset of e-values III

Sketch of Lemma 15.1's proof

Using the cdf/quantile duality (Lemma A.7 in the book: $\mathbb{P}(X > t) \le \alpha \iff q_\alpha(X) \le t$), the set $\{t \ge 1 : \mathbb{P}(E > t) \le \alpha \; \forall E \in \mathcal{E}\}$ equals $\{t \ge 1 : q_\alpha(E) \le t \; \forall E \in \mathcal{E}\}$, whose infimum is exactly $(\sup_{E \in \mathcal{E}} q_\alpha(E)) \vee 1$. A limiting ($\varepsilon \downarrow 0$) argument then shows this equals $T_\alpha(\mathcal{E})$ defined via R_γ, even though R_γ used $\ge$ rather than $>$.

Proposition 15.2: the smallest conditional e-to-p calibrator

Recall (Chapter 2) an e-to-p calibrator turns any e-value into a valid p-value. Restricted to $\mathcal{E}$, define a *conditional e-to-p calibrator* on $\mathcal{E}$ as any decreasing $f : [0, \infty] \to [0, \infty]$ such that $f(E)$ is a p-variable for every $E \in \mathcal{E}$. Then

$$x \mapsto R_{1/x}(\mathcal{E})$$

is itself a conditional e-to-p calibrator on $\mathcal{E}$, and it is the *smallest* one (i.e. $f(x) \ge R_{1/x}(\mathcal{E})$ for every other valid conditional calibrator f).

15.1 Conditional e-to-p calibrators on a subset of e-values IV

Why remarkable: on the full class $\mathfrak{E}$, the *only* useful e-to-p calibrator is $x \mapsto (1/x) \wedge 1$ (Proposition 2.4) – there is no "best" one, only this single canonical choice. Restricting to a smaller shape-constrained class $\mathcal{E}$ suddenly creates room for strictly better (uniformly smaller, hence more powerful) calibrators, and moreover a smallest one always exists.

A useful robustness remark

If $(E_\theta)_{\theta \in \Theta} \subseteq \mathcal{E}$ and E_π is any mixture $\int E_\theta \, \pi(\mathrm{d}\theta)$ over a probability measure π on Θ, then $\mathbb{P}(E_\pi \geq 1/\gamma) \leq R_\gamma(\mathcal{E})$ too. So every bound derived for $\mathcal{E}$ automatically extends to the *convex hull* of $\mathcal{E}$ – mixing over a shape-constrained family cannot break the improved threshold.

Python: R_γ and the calibrator, from a simulated e-variable (15.1)

```python
import numpy as np
rng = np.random.default_rng(0)

# A simple e-variable: E = 2*U where U ~ Uniform[0,1] under H0.
# Check validity: E[E] = 2 * 0.5 = 1  -- exactly a valid e-variable.
reps = 2_000_000
U = rng.uniform(0, 1, reps)
E = 2 * U

alpha = 0.05
naive_threshold = 1 / alpha                              # Markov threshold = 20
empirical_Rgamma = lambda z: (E >= z).mean()             # R_{1/z}({E}) empirically

# T_alpha({E}) via Lemma 15.1: sup q_alpha(E) v 1 -- here E has ONE member,
# so T_alpha is just its own (1-alpha)-quantile (floored at 1).
T_alpha = max(np.quantile(E, 1 - alpha), 1.0)
print(f"naive Markov threshold      : {naive_threshold:.3f}")
print(f"restricted threshold T_a(E) : {T_alpha:.3f}")
print(f"P(E >= T_alpha) empirically : {empirical_Rgamma(T_alpha):.4f} (target {alpha})")
# T_alpha(E) = 2 here (E is bounded by 2!) -- a HUGE improvement over 1/alpha=20,
# because knowing E's exact shape (Uniform*2) is much more informative
# than merely knowing E[E] <= 1.
```

Section 15.2 – Intuition: comonotonic e-variables

- **Comonotonic** $=$ two (or more) random variables that always move in the same direction together – like two thermometers reading the same weather, one in Celsius and one in Fahrenheit: different numbers, but always both up or both down together, because both are increasing functions of the same underlying temperature.
- If you have a whole *family* of valid e-variables (E_θ) that are comonotonic (all increasing functions of one common statistic Z), then their *supremum* $\sup_\theta E_\theta$ – which is generally NOT itself an e-variable (it's not a fixed member of the family, and its mean can exceed 1) – still satisfies the *same* Markov-type tail bound as any single member.
- Why? Because for comonotonic variables, the events $\{E_\theta \geq x\}$ are *nested* (one contains the other) rather than being spread out unpredictably – so taking a supremum over θ doesn't blow up the probability of exceeding x; it just becomes the largest of the individual (already-small) probabilities.
- **Payoff:** you can report the single best-fitting e-value from an entire comonotonic family (e.g., the maximum-likelihood member) and still control type-I error at the same α you'd get from any one fixed member – extra power, same validity.

15.2 Comonotonic e-variables I

Definition 15.3: comonotonicity

A set $\mathcal{X}_C$ of random variables is *comonotonic* if there exists a common random variable Z such that every $X \in \mathcal{X}_C$ is an increasing function of Z.

Lemma 15.4: sup of comonotonic variables preserves the tail bound

Let $x \in \mathbb{R}$, $\gamma \in [0, 1]$. If $\mathcal{X}_C$ is comonotonic and $\mathbb{P}(X \geq x) \leq \gamma$ for every $X \in \mathcal{X}_C$, then

$$\mathbb{P}\Big(\sup_{X \in \mathcal{X}_C} X \geq x \Big) \leq \gamma.$$

Step-by-step derivation of Lemma 15.4

Step 1. Every $X \in \mathcal{X}_C$ is $X = h_X(Z)$ for a common Z and increasing h_X, so $\{X \geq x\} = \{h_X(Z) \geq x\}$ is always an event of the form $\{Z \geq c_X\}$ (or $\{Z > c_X\}$) for some threshold c_X depending on X.
Step 2. Events of the form $\{Z \geq c\}$ for varying c are *totally ordered by inclusion* – for any two members X, X' of the family, either $\{Z \geq c_X\} \subseteq \{Z \geq c_{X'}\}$ or the reverse. This is exactly the "move together" property of comonotonicity.
Step 3. Therefore $\{\sup_X X \geq x\} = \bigcup_{X \in \mathcal{X}_C} \{X \geq x\}$ is a union of nested events, so its probability equals the *supremum* of the individual probabilities (a nested union's measure is the sup of the measures, not a sum):

$$\mathbb{P}\Big(\sup_{X \in \mathcal{X}_C} X \geq x\Big) = \sup_{X \in \mathcal{X}_C} \mathbb{P}(X \geq x) \leq \gamma.$$

Contrast: for a general (non-comonotonic) family, a union bound would only give $\mathbb{P}(\sup X \geq x) \leq \sum_X \mathbb{P}(X \geq x)$, which can badly exceed γ – comonotonicity is precisely what lets us avoid paying a multiple-testing penalty.

15.2 Comonotonic e-variables III

Proposition 15.5: improved threshold for comonotonic e-variables

For any comonotonic collection $\mathcal{E} \subseteq \mathfrak{E}$ of e-variables,

$$\mathbb{P}\Big(\sup_{E \in \mathcal{E}} E \geq T_\alpha(\mathcal{E})\Big) \leq \alpha, \qquad \text{in particular} \qquad \mathbb{P}\Big(\sup_{E \in \mathcal{E}} E \geq 1/\alpha\Big) \leq \alpha.$$

So $\big(\sup_{E\in\mathcal{E}} E\big)^{-1}$ is a valid p-variable, even though $\sup_{E\in\mathcal{E}} E$ is generally *not itself* a valid e-variable ($\mathbb{E}^{\mathbb{P}}[\sup_E E]$ can exceed 1).

Example 15.6: testing a normal mean

Test $\mathbb{P} = N(0,1)$ vs. $\{N(\mu,1) : \mu > 0\}$ from n iid observations $X_1, \ldots, X_n$, $S_n = \sum_i X_i$. The likelihood-ratio e-variable for a *fixed* candidate $\mu > 0$ is

$$E_\mu = \exp(\mu S_n - n\mu^2/2), \tag{Eq. 15.1}$$

and $(E_\mu)_{\mu>0}$ is comonotonic: each $E_\mu = h_\mu(S_n)$ is an increasing function of the common statistic $Z = S_n$ (whenever $\mu > 0$). **Deriving the supremum:** maximize $g(\mu) = \mu S_n - n\mu^2/2$ over $\mu > 0$. Setting $g'(\mu) = S_n - n\mu = 0$ gives the unconstrained maximizer $\mu^\star = S_n/n$; if $S_n > 0$ this is attained inside $(0, \infty)$ with value $S_n^2/(2n)$, while if $S_n \leq 0$ the supremum over $\mu > 0$ is approached as $\mu \downarrow 0$, giving value 0. Combining both cases,

$$E := \sup_{\mu>0} E_\mu = \exp\left(\frac{(S_n)_+^2}{2n}\right), \qquad (S_n)_+ := \max(S_n, 0).$$

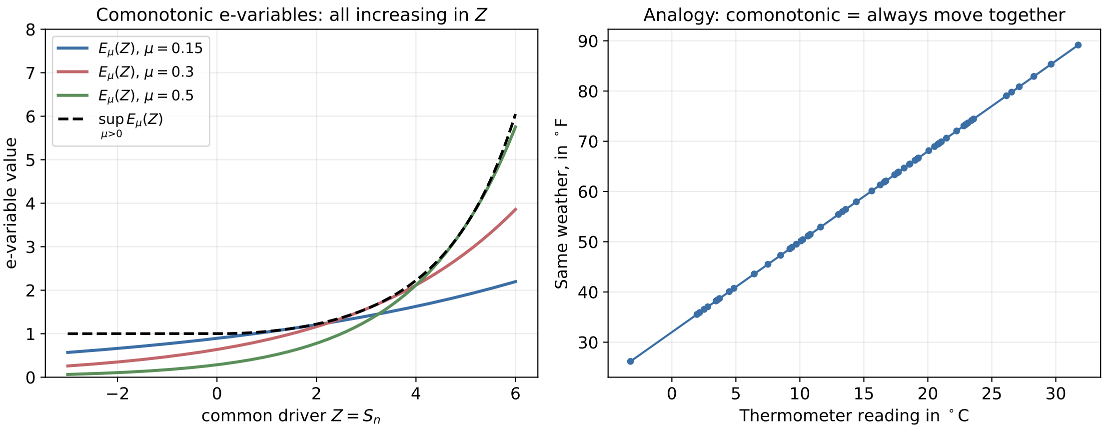
Although E is *not* itself an e-variable, Proposition 15.5 guarantees $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$ – we get to use the maximum-likelihood-flavored statistic and still control type-I error exactly at α.

Beyond likelihood ratios

Python: comonotonic e-variables and their sup (15.2)

```
import numpy as np
rng = np.random.default_rng(1)

# Testing H0: N(0,1) vs H1: N(mu,1), mu > 0, from n iid observations.
n = 20
reps = 500_000
X = rng.normal(0, 1, size=(reps, n))          # data simulated UNDER H0
S_n = X.sum(axis=1)

def E_mu(mu, S):
    return np.exp(mu * S - n * mu**2 / 2)

# A grid approximating the comonotonic family (E_mu)_{mu > 0}:
mu_grid = np.linspace(0.01, 3.0, 400)
E_family = np.stack([E_mu(mu, S_n) for mu in mu_grid], axis=1)   # (reps, |grid|)
E_sup_grid = E_family.max(axis=1)                                  # sup over mu>0

# Closed form: sup_{mu>0} E_mu = exp( (S_n)_+^2 / (2n) )
E_sup_closed = np.exp(np.clip(S_n, 0, None)**2 / (2 * n))

alpha = 0.05
print("max |grid sup - closed form| :", np.abs(E_sup_grid - E_sup_closed).max())
print(f"P(sup E_mu >= 1/alpha) empirically : {(E_sup_closed >= 1/alpha).mean():.4f}"
      f"  (must be <= alpha = {alpha})")
# Confirms Proposition 15.5: even though E_sup is not a valid e-variable
# (check E[E_sup] > 1 below!), thresholding it at 1/alpha is still valid.
print("E[E_sup] =", E_sup_closed.mean(), " (> 1, so NOT itself an e-variable)")
```

Figure: comonotonic e-variables, illustrated



Left: the family $E_\mu(Z) = \exp(\mu Z - n\mu^2/2)$ for several $\mu > 0$, all increasing in the common driver $Z = S_n$, with envelope $\sup_{\mu>0} E_\mu(Z)$ (Example 15.6). Right: the thermometer analogy – two rescalings of the same underlying quantity always move together.

Section 15.3 – Intuition: density conditions

- A **density condition** is an assumption that the e-value's distribution has a well-behaved density – ruling out the pathological, spiky/atomic worst-case distributions that make the plain Markov bound tight.
- Recall: the worst-case e-variable for Markov's inequality puts *all* its mass at exactly two points (0 and $1/\alpha$) – extremely spiky. If we know E has, say, a *decreasing* density (most mass concentrated near small values, smoothly tapering off, as with an Exponential or Pareto variable), that spiky worst case is structurally impossible.
- **Punch line of this section:** a decreasing (or unimodal) density on the e-value lets us cut the rejection threshold roughly *in half*: instead of $1/\alpha$, we can reject at about $1/(2\alpha)$ – a substantial, checkable power gain, for a mild and often plausible shape assumption.

15.3 Density conditions I

Three shape-constrained classes

$$\mathcal{E}_{\mathrm{D}} = \{E \in \mathfrak{E} : E \text{ has a decreasing density on its support}\},$$

$$\mathcal{E}_{\mathrm{D}>1} = \{E \in \mathfrak{E} : E \text{ has a decreasing density over } [1, \infty)\},$$

$$\mathcal{E}_{\mathrm{U}} = \{E \in \mathfrak{E} : E \text{ has a unimodal density on } [0, \infty)\}.$$

A density on $[0, \infty)$ is *unimodal* if it increases on $[0, x^\star]$ and decreases on $[x^\star, \infty)$ for some mode $x^\star \geq 0$ (point masses at the mode are allowed). Note $\mathcal{E}_{\mathrm{D}} \subseteq \mathcal{E}_{\mathrm{U}}$, so $R_\gamma(\mathcal{E}_{\mathrm{U}}) \geq R_\gamma(\mathcal{E}_{\mathrm{D}})$. Decreasing densities cover, e.g., Exponential and Pareto e-variables; unimodal densities additionally cover Log-normal and Gamma. These are conditions on the e-value's *own* distribution, not on the hypotheses being tested.

15.3 Density conditions II

Theorem 15.7: the improved bounds

For $\gamma \in (0,1]$:

(i) $R_\gamma(\mathcal{E}_{\mathrm{D}}) = \gamma/2$ for $\gamma \neq 1$, and $R_1(\mathcal{E}_{\mathrm{D}}) = 1$;

(ii) $R_\gamma(\mathcal{E}_{\mathrm{D}>1}) = \gamma\big/\big(1 + \sqrt{1-\gamma^2}\big)$;

(iii) $R_\gamma(\mathcal{E}_{\mathrm{U}}) = (\gamma/2) \vee (2\gamma - 1)$.

Step-by-step derivation of $R_\gamma(\mathcal{E}_{\mathrm{D}}) = \gamma/2$ (part (i))

Fix $z > 1$ and set $\gamma = 1/z$ at the end. We seek $\sup_{E \in \mathcal{E}_{\mathrm{D}}} \mathbb{P}(E \geq z)$ subject to $\mathbb{E}^{\mathbb{P}}[E] = 1$ (validity) and a decreasing density.

Step 1 – reduce to a two-parameter extremal family. Among all decreasing-density E with left endpoint of support at 0, the one that maximizes $\mathbb{P}(E \geq z)$ for fixed mean 1 turns out to be a mixture of a point mass at 0 (weight $1 - w$) and a Uniform$[0, b]$ density (weight w), for some $b \geq z$. (Intuition: moving density mass *away* from the middle, either down to 0 or spread thin near the tail cutoff z, only helps push more mass past z while respecting "decreasing density.")

Step 2 – impose the mean-1 constraint. The mean of this mixture is $w \cdot (b/2)$ (point mass contributes 0), so $\mathbb{E}[E] = 1$ forces $w = 2/b$.

Step 3 – write down $\mathbb{P}(E \geq z)$ and optimize over b.

$$\mathbb{P}(E \geq z) = w \cdot \frac{b - z}{b} = \frac{2}{b} \cdot \frac{b - z}{b} = \frac{2(b - z)}{b^2}.$$

Differentiate in b and set to 0: $\dfrac{\mathrm{d}}{\mathrm{d}b}\left[\dfrac{2(b-z)}{b^2}\right] = \dfrac{2(2z - b)}{b^3} = 0 \implies b^\star = 2z.$

Step 4 – plug back in. At $b^\star = 2z$: $w^\star = 2/(2z) = 1/z$, and

Figure: worst-case type-I error under density conditions

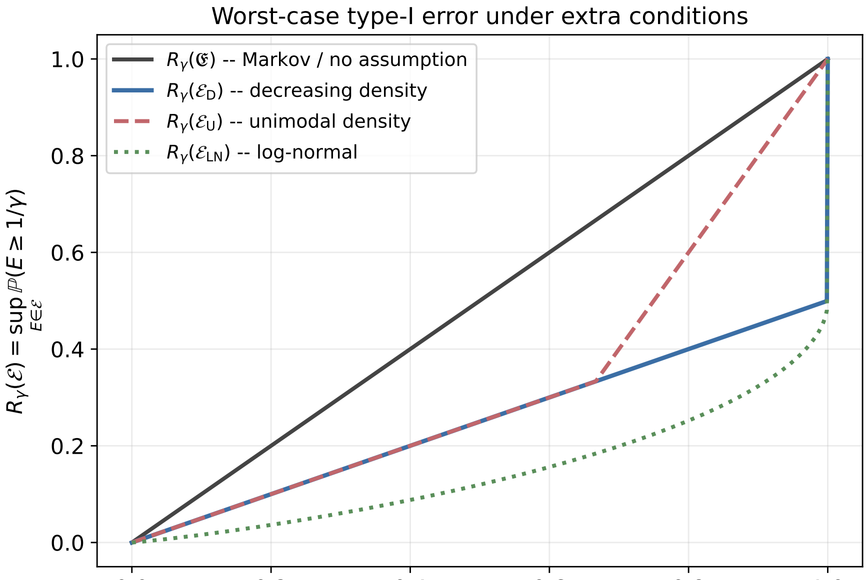
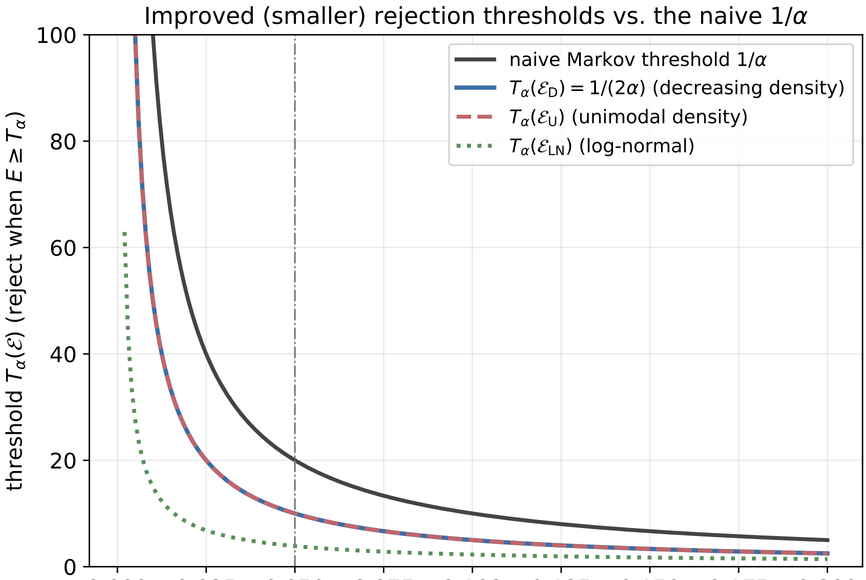


Figure: the resulting improved thresholds



Python: verifying the decreasing-density bound (15.3)

```python
import numpy as np
rng = np.random.default_rng(2)

alpha = 0.05
naive_T = 1 / alpha                    # = 20
improved_T = 1 / (2 * alpha)           # = 10, Theorem 15.7(i)

reps = 3_000_000

# Two DIFFERENT decreasing-density e-variables in E_D, both with mean 1:
# (a) Exponential(1): density e^{-x} on [0, infty), strictly decreasing.
E_exp = rng.exponential(scale=1.0, size=reps)

# (b) Pareto-type: point mass 1-w at 0, Uniform[0,b] w.p. w (the EXTREMAL
#     family from the Theorem 15.7(i) derivation), with b = 2*improved_T,
#     w = 1/improved_T  -- this is the worst case, so it should ATTAIN gamma/2.
b = 2 * improved_T
w = 1 / improved_T
mask = rng.uniform(0, 1, reps) < w
E_extremal = np.where(mask, rng.uniform(0, b, reps), 0.0)

for name, E in [("Exponential(1)", E_exp), ("Extremal mixture", E_extremal)]:
    p_naive = (E >= naive_T).mean()
    p_improved = (E >= improved_T).mean()
    print(f"{name:18s}: P(E>=1/alpha)={p_naive:.5f}  P(E>=1/(2alpha))={p_improved:.5f}"
          f"  (target R_gamma(E_D) = alpha/2 = {alpha/2:.5f})")
# Both respect P(E >= 1/(2 alpha)) <= alpha/2; the extremal mixture attains it with equality.
```

Running example, part 1: setup (coin-fairness style)

Recall the coin-fairness running example (cf. Chapter 3)

Coin flips $X_i \in \{-1, +1\}$ ($X_i = +1$ for heads with probability p), so $\mathbb{E}[X_i] = 2p - 1 =: \mu$. Under H_0 ($p = 1/2$, fair coin), X_i has mean 0 and variance 1. By the CLT, for moderately large n the sum $S_n = \sum_{i=1}^{n} X_i$ is approximately $N(0, n)$ – so the Gaussian likelihood-ratio e-variable of Example 15.6,

$$E_\mu = \exp\left(\mu S_n - n\mu^2/2\right),$$

is (approximately, via the CLT) a sequential e-variable for detecting a biased coin with bias $\mu = 2p - 1$.

A key structural fact: E_μ is exactly log-normal under H_0

Under H_0, $S_n \sim N(0, n)$ (in the Gaussian/CLT approximation), so

$$\log E_\mu = \mu S_n - n\mu^2/2 \sim N\left(-n\mu^2/2,\, n\mu^2\right).$$

That is, E_μ has a *log-normal* distribution: $E_\mu \in \mathcal{E}_{\mathrm{LN}}$ (Section 15.4). Since log-normal densities are unimodal, also $E_\mu \in \mathcal{E}_{\mathrm{U}}$ (Section 15.3). This single e-variable is thus a perfect vehicle for the improved thresholds of *both* sections.

Running example, part 2: improved thresholds & how much power they buy

Step 1: the improved thresholds at $\alpha = 0.05$

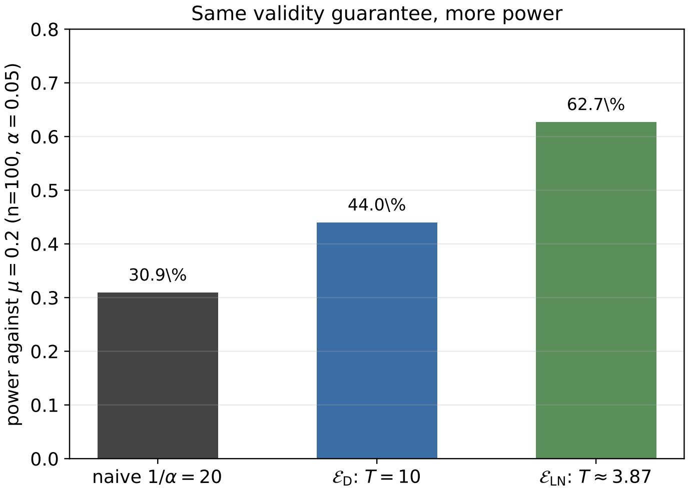
- Decreasing/unimodal density (Theorem 15.7): $T_\alpha(\mathcal{E}_{\mathrm{D}}) = 1/(2\alpha) = \mathbf{10}$ – exactly half the naive threshold.
- Log-normal (Theorem 15.10(v), since $R_\gamma(\mathcal{E}_{\mathrm{LN}}) = \Phi(-\sqrt{-2\log\gamma})$): solving $\Phi(-\sqrt{-2\log\gamma}) = 0.05$ numerically gives $\gamma^\star \approx 0.2585$, so $T_\alpha(\mathcal{E}_{\mathrm{LN}}) = 1/\gamma^\star \approx \mathbf{3.87}$ – more than $5\times$ smaller than naive!

Step 2: what this buys in power (fixed $\mu = 0.2$, i.e. $p = 0.6$, $n = 100$ flips)

Reject when $S_n \geq S^\star(T) := \log(T)/\mu + n\mu/2$. Under the true bias $\mu = 0.2$, $S_n \sim N(n\mu, n) = N(20, 100)$ (CLT approx.), so power $= 1 - \Phi\big((S^\star(T) - 20)/10\big)$:

Threshold used	T	critical $S^\star(T)$	power against $\mu = 0.2$
naive Markov $(1/\alpha)$	20	24.98	30.9%
decreasing density $\mathcal{E}_{\mathrm{D}}$	10	21.51	44.0%
log-normal $\mathcal{E}_{\mathrm{LN}}$	3.87	16.76	62.7%

Figure: the power gain, visualized



Power against $\mu = 0.2$ ($n = 100$, $\alpha = 0.05$) using the naive threshold vs. the improved thresholds from the density condition ($\mathcal{E}_{\mathrm{D}}$) and the log-normal condition ($\mathcal{E}_{\mathrm{LN}}$). Same validity guarantee throughout – only the

Section 15.4 – Intuition: why look at $\log E$ instead of E?

- Many e-values used in practice are *products* of many small per-observation e-values (e-processes, Chapter 7). A product's *logarithm* is a *sum* – and sums of many small, roughly-independent pieces tend towards Gaussian-like shapes by the Central Limit Theorem.
- So instead of assuming a density condition directly on E (Section 15.3), it is often more natural/plausible to assume a shape condition on $\log E$: e.g., that $\log E$ is symmetric, unimodal, or itself has a decreasing density.
- **Surprising subtlety:** not every such assumption on $\log E$ actually helps! Assuming $\log E$ is merely *symmetric*, or merely *unimodal* (without more), turns out to give *no improvement at all* over the naive $1/\alpha$ threshold, in the practically relevant range. You need slightly more structure (decreasing density of $\log E$, or both unimodal *and* symmetric, or exactly log-normal) to see a real gain.

15.4 Conditions on log-transformed e-variables I

Six classes (all require $\mathbb{P}(E=0)=0$ so that $\log E$ is real-valued)

$$\mathcal{E}_{\mathrm{LS}} = \{E : \log E \text{ symmetric}\}, \quad \mathcal{E}_{\mathrm{LU}} = \{E : \log E \text{ unimodal density}\},$$

$$\mathcal{E}_{\mathrm{LD}>0} = \{E : \log E \text{ decreasing density on } [0,\infty)\}, \quad \mathcal{E}_{\mathrm{LD}} = \{E : \log E \text{ decreasing density}\},$$

$$\mathcal{E}_{\mathrm{LUS}} = \{E : \log E \text{ unimodal } \textit{and} \text{ symmetric}\}, \quad \mathcal{E}_{\mathrm{LN}} = \{E : E \text{ log-normal}\}.$$

A random variable X is *symmetric* if $X \overset{d}{=} c - X$ for some constant c. Containments: $\mathcal{E}_{\mathrm{LN}} \subseteq \mathcal{E}_{\mathrm{LUS}} \subseteq \mathcal{E}_{\mathrm{LS}}$ and $\mathcal{E}_{\mathrm{LD}} \subseteq \mathcal{E}_{\mathrm{LD}>0}$. ($E \in \mathfrak{E}$ is log-normal iff $\log E \sim N(\mu, \sigma^2)$ with $\mu + \sigma^2/2 \leq 0$, the constraint needed for $\mathbb{E}[E] \leq 1$.)

Theorem 15.10: the improved bounds

For $\gamma \in (0,1]$:

(i) $R_\gamma(\mathcal{E}_{\mathrm{LS}}) = \gamma \wedge (1/2)$ for $\gamma \neq 1$ $(R_1 = 1)$ – **no improvement** for $\gamma \leq 1/2$ (i.e., essentially always in practice);

(ii) $R_\gamma(\mathcal{E}_{\mathrm{LU}}) = \gamma$ – **no improvement at all**, ever;

(iii) $\gamma/(\mathrm{e}-\gamma) \leq R_\gamma(\mathcal{E}_{\mathrm{LD}>0}) = \mathrm{e}^{-a_\gamma} \leq \gamma/(1+\sqrt{1-\gamma^2})$, where a_γ solves $\mathrm{e}^a(1-a-\log\gamma) = 1$;

(iv) $R_\gamma(\mathcal{E}_{\mathrm{LD}}) = R_\gamma(\mathcal{E}_{\mathrm{LUS}})$, and $\gamma/\mathrm{e} \leq R_\gamma(\mathcal{E}_{\mathrm{LD}}) \leq \big(\gamma/(\mathrm{e}(1-\gamma^2))\big) \wedge \big(\gamma/(1+\sqrt{1-\gamma^2})\big)$;

(v) $R_\gamma(\mathcal{E}_{\mathrm{LN}}) = \Phi\big(-\sqrt{-2\log\gamma}\big) \leq \gamma/\big(2\sqrt{-\pi\log\gamma}\big)$, for $\gamma \neq 1$ $(R_1 = 1)$; Φ is the standard normal cdf.

15.4 Conditions on log-transformed e-variables III

Why (i) and (ii) are cautionary tales

Symmetry or unimodality of $\log E$ *alone* says nothing about how thin the tail is – $\log E$ could still have an extremely heavy (or even non-decreasing near its mode, spread-out) tail consistent with symmetry/unimodality, reproducing the worst-case Markov scenario. **Lesson:** vague shape words are not enough; you need an assumption that actually constrains the tail, such as a genuinely *decreasing* density (of $\log E$ or of E itself), or the fully parametric log-normal assumption.

Why (iii)–(v) do help: roughly a factor of e ≈ 2.718

For small γ, $R_\gamma(\mathcal{E}_{\mathrm{LD}>0})$, $R_\gamma(\mathcal{E}_{\mathrm{LD}}) = R_\gamma(\mathcal{E}_{\mathrm{LUS}})$ all behave like γ/e – i.e., $T_\alpha(\mathcal{E}) \approx \mathrm{e}/\alpha$ is *wrong*; correctly inverting, the threshold shrinks by a factor of roughly e relative to $1/\alpha$ for small α. The *log-normal* assumption (v) is the strongest of all: $R_\gamma(\mathcal{E}_{\mathrm{LN}}) \to 0$ much faster than linearly in γ as $\gamma \to 0$ (note the $\sqrt{-\log\gamma}$ inside Φ), giving by far the smallest threshold of every class in this chapter.

Connecting back to Section 15.3

$\mathcal{E}_{\mathrm{LD}>0} \subseteq \mathcal{E}_{\mathrm{D}>1}$ (a decreasing-density-of-log condition on $[0, \infty)$ implies the raw variable has a decreasing density on $[1, \infty)$, since $\log E \geq 0 \iff E \geq 1$) – so the two sections' results are genuinely compatible refinements of each other, not competing theories.

Table 15.12: improved thresholds $T_\alpha(\mathcal{E})$ (reproduced)

	α					
Class	0.001	0.01	0.02	0.05	0.1	0.2
$\mathfrak{E},\ \mathcal{E}_{\mathrm{LS}}$	1000	100	50	20	10	5
$\mathcal{E}_{\mathrm{D}},\ \mathcal{E}_{\mathrm{U}}$	500	50	25	10	5	2.5
$\mathcal{E}_{\mathrm{D}>1}$	500	50.01	25.01	10.03	5.05	2.60
$\mathcal{E}_{\mathrm{LUS}},\ \mathcal{E}_{\mathrm{LD}}$	367.88	36.82	18.45	7.49	3.93	2.25
$\mathcal{E}_{\mathrm{LD}>0}$	368.25	37.16	18.77	7.73	4.07	2.25
$\mathcal{E}_{\mathrm{LN}}$	118	14.97	8.24	3.87	2.27	1.42

Reading the $\alpha = 0.05$ column: naive 20 (row 1) $\to$ 10 (density condition) $\to$ 7.49 (log-transform, decreasing/unimodal-symmetric) $\to$ 3.87 (log-normal) – each additional, checkable shape assumption buys a further reduction in the rejection threshold, i.e. more power at the same validity level.

Python: log-transform thresholds and Table 15.12 (15.4)

```python
import numpy as np
from scipy.stats import norm
from scipy.optimize import brentq

def R_gamma_LN(gamma):
    """Theorem 15.10(v): R_gamma(E_LN) = Phi(-sqrt(-2 log gamma))."""
    return norm.cdf(-np.sqrt(-2 * np.log(gamma)))

def T_alpha_LN(alpha):
    """Invert R_gamma(E_LN) = alpha for gamma, then T = 1/gamma."""
    gamma_star = brentq(lambda g: R_gamma_LN(g) - alpha, 1e-9, 1 - 1e-9)
    return 1 / gamma_star

for alpha in [0.001, 0.01, 0.02, 0.05, 0.1, 0.2]:
    naive = 1 / alpha
    T_LN = T_alpha_LN(alpha)
    print(f"alpha={alpha:<6} naive=1/alpha={naive:8.2f}   T_alpha(E_LN)={T_LN:7.3f}"
          f"   improvement factor = {naive / T_LN:5.2f}x")
# Reproduces Table 15.12's E_LN row, e.g. alpha=0.05 -> T ~ 3.87 (a ~5.2x
# reduction versus the naive Markov threshold of 20).
```

Chapter 15 summary

- The plain Markov threshold $1/\alpha$ (Chapter 2) is tight only against a pathological, spiky worst-case e-variable. Restricting to a shape-constrained class $\mathcal{E} \subsetneq \mathfrak{E}$ lets us compute a strictly smaller, valid threshold $T_\alpha(\mathcal{E}) \leq 1/\alpha$ (Section 15.1, Lemma 15.1).
- **Comonotonicity** (15.2): if a family of e-variables all move together (increasing functions of one common Z), their *supremum* obeys the same tail bound as any single member (Lemma 15.4, Proposition 15.5) – letting you use the best-fitting member of a parametric family (e.g., an MLE-based likelihood ratio) at no cost to validity.
- **Density conditions** (15.3): a decreasing or unimodal density on E itself roughly *halves* the threshold ($T_\alpha(\mathcal{E}_{\mathrm{D}}) = 1/(2\alpha)$, Theorem 15.7) – derived via an explicit worst-case mixture (point mass $+$ uniform tail).
- **Log-transform conditions** (15.4): shape assumptions on $\log E$ (motivated by products of e-values / CLT) can shrink the threshold by roughly a factor of e (decreasing density, or unimodal$+$symmetric) or dramatically more (exactly log-normal) – but symmetry or unimodality of $\log E$ *alone* gives no improvement (Theorem 15.10).
- **Running numeral:** testing a coin's bias ($\mu = 0.2$, $n = 100$, $\alpha = 0.05$), the naive threshold 20 gives power 30.9%; the density-condition threshold 10 gives power 44.0%; the log-normal threshold 3.87 gives power 62.7% – all with exactly the same $\alpha = 0.05$ type-I error guarantee.

Bibliographical note

- This chapter is largely based on Blier-Wong and Wang [2024], where full proofs of all results (including the omitted proofs of Theorems 15.7(ii)–(iii) and 15.10) can be found, along with extensions to combined e-values and the e-BH procedure.
- General background on probability bounds under distributional shape information: Shaked and Shantikumar [2007], Rüschendorf et al. [2024].
- Comonotonicity (Section 15.2) is a central concept in dependence modeling and in decision theory under ambiguity, via its connection to Choquet integrals [Schmeidler, 1989]. A standard reference is Denneberg [1994].